

Lecture 11 Live Problems

Fitting models of growth: linear, power, exponential.

Points toward lab manual pages: **Fit1, Fit2, Fit3, Fit4** (section L2 S5)

1. The bathtub. You close the drain in your tub, put in some lavender bubble bath, and turn the faucet until hot water comes out. Let $V(t)$ be the volume of water in the tub.

- (a) What are appropriate units?
- (b) What is the appropriate sort of model for this situation?
- (c) What is $V(0)$? What is the change equation?
- (d) What is a reasonable rate of change?

2. The water balloon. Water is entering a water balloon at a steady flow rate. The water balloon gets bigger! Let $R(t)$ be the radius of the water balloon.

- (a) What are appropriate units?
- (b) What is the appropriate sort of model for this situation?
- (c) Can you guess some of the parameters, just from the situation?

3. Fruit flies. A population of fruit flies (*Drosophila*) is composed of males and females. Both live for about 30 days. A typical female lays about 700 eggs in her lifetime. Let $P(t)$ be the number of fruit flies in a large container with plenty of food.

- (a) What is the appropriate sort of model for this situation?
- (b) Give an appropriate change equation, with plausible parameters.